

POD-RBF reduced-order modeling for phase-change processes in fluid dynamics

Youssef Nasser^{*}

*Institut de Mécanique et d'Ingénierie de Bordeaux,
Université de Bordeaux, CNRS, Bordeaux INP
Talence, France*

Abstract

This paper investigates a non-intrusive reduced-order modeling approach to numerically simulate phase-change material with varying regimes. Instead of the Galerkin projection approach, the method discussed here is based on radial basis functions (RBF) combined with proper orthogonal decomposition (POD). The POD component provides a spatial approximation with a reduced degree of freedoms, while the RBF module models the temporal evolution by interpolating the POD coefficients. The resulting POD–RBF method yields an accurate, global, nonlinear interpolation in real time and is essentially mesh-free. To enhance consistency with physical constraints, the model is further optimized through an *a posteriori* error estimator that quantifies both interpolation and residual errors. To assess the effectiveness of the proposed methodology, two configurations are examined: first, a purely conductive regime governed by the heat transfer equation, and second, a convective regime described by the Navier–Stokes–Boussinesq equations.

Keywords: Reduced-order modeling; Proper orthogonal decomposition; Radial basis functions; Finite element method; a *posteriori* error estimation; Greedy algorithms; Phase-change materials.

Contents

1	Introduction	2
2	Learning methodology	4
3	Adaptive basis enrichment	5
4	Numerical results	6
4.1	PCM in conductive regime	6
4.2	PCM in convective regime	9
5	Conclusion	13
A	POD-Galerkin model	14
B	POD and PIPOD algorithms	15

^{*}youssef.nasser@u-bordeaux.fr

1 Introduction

Phase-change materials (PCMs) play a crucial role in a wide range of thermal engineering applications, including energy storage systems, thermal regulation, and heat transfer enhancement [ZMCM03, FKRAH04, STCB09]. Melting and solidification processes have been extensively studied using numerical methods in both conductive and convective regimes, and they are commonly modeled using enthalpy-based formulations coupled with the Navier–Stokes–Boussinesq equations [VC87, RDD19, CSZX21]. These processes are characterized by strong nonlinearities arising from latent heat effects and, in many practical situations, from the tight coupling between thermal and fluid flow phenomena.

In this work, we first consider a PCM process in a purely conductive regime where convective effects are neglected. In this case, the temperature evolution is governed by the following heat transfer problem:

$$\partial_t T - \frac{1}{\text{Re Pr}} \nabla^2 T + \partial_t S(T) = 0, \quad \text{in } \Omega \times (0, t_{\max}), \quad (1)$$

$$T(\mathbf{x}, 0) = T_0, \quad \text{in } \Omega, \quad (2)$$

where T denotes the temperature and $S(T)$ represents the enthalpy that controls the phase-change process, defined by

$$S(T) = \begin{cases} \frac{1}{\text{Ste}}, & \text{if } T \geq T_f, \\ 0, & \text{otherwise,} \end{cases}$$

where $T_f = 0$ is the fusion temperature and $\text{Ste} = 0.045$ is the Stefan number.

More generally, a melting PCM experiencing natural convection is governed by the Navier–Stokes–Boussinesq equations [SRT+20] coupled with energy conservation:

$$\nabla \cdot \mathbf{u} = 0, \quad (3a)$$

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \frac{1}{\text{Re}} \nabla^2 \mathbf{u} = \frac{\text{Ra } T}{\text{Pr Re}^2} \mathbf{e}_y - \frac{C_{\text{ck}}(1 - \zeta_l(T))^2}{\zeta_l(T)^3 + 10^{-6}} \mathbf{u}, \quad (3b)$$

$$\partial_t T + \mathbf{u} \cdot \nabla T - \frac{1}{\text{Re Pr}} \nabla^2 T = -\partial_t S(T), \quad (3c)$$

where $\mathbf{u}$ denotes the velocity field, p is the pressure, and $\text{Ra} = 3.27 \times 10^5$, $\text{Re} = 1$, and $\text{Pr} = 56.2$, represent respectively the Rayleigh, Reynolds, Prandtl numbers. $C_{\text{ck}} = 10^6$ denotes the Carman–Kozeny penalization constant and the liquid fraction $\zeta_l(T)$ is defined as

$$\zeta_l(T) = \begin{cases} 1, & \text{if } T \geq T_f, \\ 0, & \text{otherwise,} \end{cases}$$

The temperature is prescribed as $T = \mu$ on the left boundary, where μ serves as a parametric boundary temperature, while homogeneous Neumann boundary conditions are applied on the remaining walls. Dirichlet boundary conditions are imposed for the velocity field. This formulation is widely used to model buoyancy-driven melting and solidification in enclosures [GV96, BVR88], where the Carman–Kozeny penalization technique damps the velocity field within the solid region [Vol90].

High-fidelity finite element methods are commonly employed to numerically solve PCM problems due to their ability to accurately capture coupled heat transfer, convective flows, and moving solid–liquid interfaces in complex geometries [RDD19, SRT+20, VKS25, YA24]. Problems (1) and (3) are approximated on unstructured squared meshes following the methodology presented in [SRT+20]. In this approach, the temperature T , velocity $\mathbf{u} = (u_1, u_2)$, and pressure p are projected onto their

respective finite element spaces V_h^T , V_h^u , and W_h as follows:

$$\begin{aligned}
T_h(\mathbf{x}, t) &= \sum_{i=1}^{N_h^T} a_i(t) v_i(\mathbf{x}), & v_i &\in V_h^T, \\
u_{h,j}(\mathbf{x}, t) &= \sum_{i=1}^{N_h^u} b_{i,j}(t) v_i(\mathbf{x}), & v_i &\in V_h^u, \quad j = 1, 2, \\
p_h(\mathbf{x}, t) &= \sum_{i=1}^{N_h^p} c_i(t) q_i(\mathbf{x}), & q_i &\in W_h,
\end{aligned} \tag{4}$$

where $a_i(t)$, $b_{i,j}(t)$, and $c_i(t)$ represent the time-dependent degrees of freedom. By substituting these expansions into the governing equations and applying a standard Galerkin projection, we obtain a nonlinear system of algebraic-differential equations for the temporal coefficients. Due to the strong nonlinearity of the governing equations and the discontinuous nature of the phase-change terms principally the enthalpy, the use of high-order finite elements for spatial discretization, combined with robust nonlinear time-integration schemes, is essential to achieve accurate solutions. These requirements lead to high computational costs, particularly when simulations must be performed across multiple parameter values or over long time horizons. In such contexts, reduced-order modeling (ROM) techniques provide an efficient alternative by capturing the dominant dynamics of the full-order system within a significantly lower-dimensional space [QMN16, BGW15]. Similar to the high-fidelity discretization, the reduced temperature T_r , velocity $\mathbf{u}_r = (u_{r,1}, u_{r,2})$, and pressure p_r are approximated by projecting them onto their respective low-dimensional POD spaces $V_r^T \subset V_h$, $V_r^u \subset V_h^u$, and $W_r \subset W_h$ as follows:

$$\begin{aligned}
T_r(\mathbf{x}, t) &= \sum_{i=1}^{r^T} a_i(t) \phi_i^T(\mathbf{x}), & \phi_i^T &\in V_r^T, \\
u_{r,j}(\mathbf{x}, t) &= \sum_{i=1}^{r^u} b_{i,j}(t) \phi_i^u(\mathbf{x}), & \phi_i^u &\in V_r^u, \quad j = 1, 2, \\
p_r(\mathbf{x}, t) &= \sum_{i=1}^{r^p} c_i(t) \psi_i(\mathbf{x}), & \psi_i &\in W_r,
\end{aligned} \tag{5}$$

where ϕ_i^T , ϕ_i^u , and ψ_i represent the spatial POD modes extracting from high-fidelity snapshot data. The reduced temporal coefficients $a_i(t)$, $b_{i,j}(t)$, and $c_i(t)$ are determined as previously through Galerkin projection [SSB⁺21, QMN16, HRS16, HLB96, KV01, Vol11]. While this intrusive strategy often provides high accuracy and preserves the underlying physics, it requires explicit access to the full-order operators and typically involves expensive online evaluations for nonlinear and parameter-dependent problems. For instance, in the temperature equation, a second-order time integration scheme is required to properly capture the strong nonlinear dynamics associated with the phase change, especially the sharp variations induced by the enthalpy function $S(T)$. Projecting these governing equations onto a reduced basis introduces a major bottleneck: both the enthalpy term $S(T)$ and its derivative $S'(T)$ must be tracked or approximated in the reduced space. To alleviate this issue, hyper-reduction techniques such as the Discrete Empirical Interpolation Method (DEIM) [CS10] are commonly employed to approximate nonlinear terms efficiently. Nevertheless, in the present context, the highly localized and steep variations of the enthalpy function near the phase interface make these nonlinear terms exceptionally difficult to approximate accurately within a low-dimensional subspace. As a result, even with hyper-reduction, the reduced-order model may suffer from a significant loss of accuracy and stability. To balance the high computational overhead of the intrusive POD-Galerkin method against the potential inaccuracies of DEIM-based hyper-reduction, Radial Basis Function (RBF) interpolation offers a compelling alternative that provides fast, non-intrusive, and accurate approximations.

Rather than projecting the governing equations, the POD-RBF approach constructs a non-intrusive surrogate model that directly maps the parametric and temporal dependencies to the reduced coefficients learned from data [Buh03]. This strategy bypasses the need to explicitly evaluate or hyper-reduce the discontinuous functions $S(T)$ and $S'(T)$, thereby avoiding the associated projection and interpolation errors. Moreover, because the solution manifold remains relatively smooth despite the sharp

nonlinearities in the continuous equations, RBF interpolation can effectively capture the global system dynamics using a limited number of training snapshots. Consequently, non-intrusive POD-RBF strategies have recently emerged as powerful alternatives for parametric and time-dependent problems [XFPH15, HRS18, GH18, WA19]. Despite these advantages, the accuracy of standard POD-RBF models can degrade due to interpolation errors and a lack of guaranteed physical consistency. To address these limitations, this work proposes an enhanced POD-RBF framework incorporating an *a posteriori* error estimator and a Greedy sampling strategy to adaptively improve model accuracy, building upon recent advances in adaptive, data-driven ROMs [WACRX25, FCB24, MPR15, AGZ⁺15, PW16]. The capabilities of the proposed approach are demonstrated on both conductive and convective PCM melting benchmarks.

The remainder of this paper is organized as follows. Section 2 describes the core POD-RBF methodology, while adaptive basis enrichment based on the Greedy procedure is detailed in Section 3. Numerical verification and results are presented in Section 4, focusing first on the conductive regime (Section 4.1) and subsequently on the convective Navier-Stokes regime (Section 4.2).

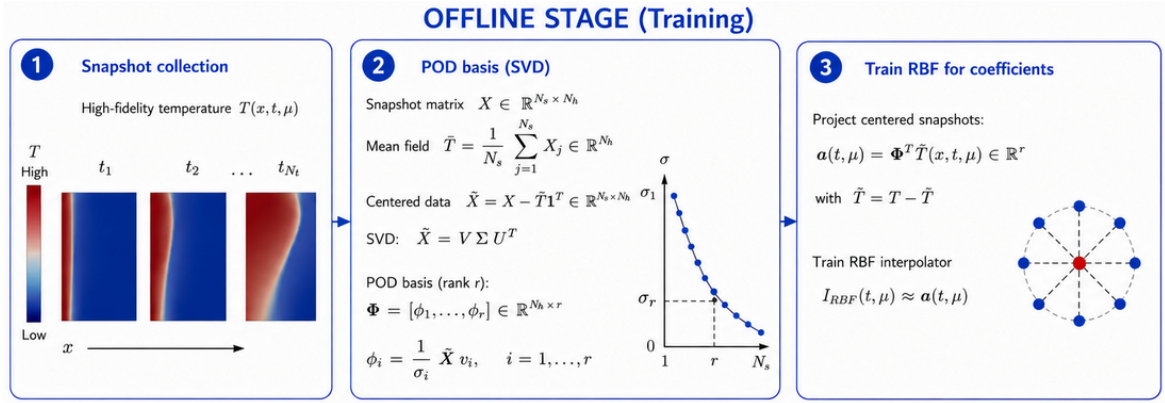


Figure 1: Schematic of the non-intrusive POD-RBF reduced-order modeling. Offline, POD modes and reduced coefficients are extracted from snapshots.

2 Learning methodology

Suppose we have a set of data snapshots $[T_h^1, T_h^2, \dots, T_h^{N_s}]$ collected at different parameter instances (t^n, μ^n) , for $n = 1, \dots, N_s$, where $T_h^n \in \mathbb{R}^{N_h}$ represents a numerical scalar field, with N_h being the number of degrees of freedom. In the following, as already proposed in [XFPH15], the combined POD-RBF framework is employed to construct a surrogate model in two distinct stages: offline and online. During the offline stage, the reduced basis and the kernel interpolator are constructed from the snapshot data, as illustrated in Figure 1. The RBF interpolator $\mathcal{I}_{\text{RBF}}$ is explicitly defined as:

$$\mathcal{I}_{\text{RBF}}^i(t, \mu) = \sum_{m=1}^{N_s} w_{i,m} \varphi(||(t, \mu) - (t_m, \mu_m)||), \quad i = 1, \dots, r, \quad (6)$$

where the radial function φ may take one of the forms:

$$\text{Cubic: } \varphi(||(t, \mu)||) = ||(t, \mu)||^3, \quad \text{Spline: } \varphi(||(t, \mu)||) = ||(t, \mu)||^2 \log(||(t, \mu)||).$$

The RBF weights $w_{i,m}$ are computed offline by enforcing the interpolation condition:

$$\mathcal{I}_{\text{RBF}}^i(t, \mu) = \phi_i(\mathbf{x}) \tilde{T}(\mathbf{x}, t, \mu), \quad i = 1, \dots, N_s.$$

The dimension r of the reduced basis Φ is determined according to the cumulative energy criterion:

$$\frac{\sum_{k=1}^r \sigma_k^2}{\sum_{k=1}^{N_s} \sigma_k^2} \geq 1 - \epsilon,$$

with $\sigma_{k=1}^{N_s}$ denoting the singular values of the snapshot matrix. The tolerance ϵ is selected as small as possible in order to guarantee sufficient approximation accuracy while maintaining a compact reduced basis. Subsequently, during the online stage, the resulting RBF model is evaluated at any new query point (t^*, μ^*) to reconstruct the reduced solution:

$$T_r(\mathbf{x}, t^*, \mu^*) = \bar{T}(\mathbf{x}) + \sum_{i=1}^r \phi_i(\mathbf{x}) \mathcal{I}_{RBF}^i(t^*, \mu^*), \quad (7)$$

where $\bar{T}(\mathbf{x})$ denotes the mean of the snapshots, which serves to reinforce the boundary conditions. The implementation of this classical POD method as well as pseudo-incremental POD (PIPOD) variant are revised in Appendix B.

3 Adaptive basis enrichment

The POD-RBF framework presented above is fully non-intrusive, as it does not require explicit access to the governing partial differential equations. However physical constraints and mesh geometric metrics are needed to enhance the accuracy and robustness of the surrogate model. To compromise the lack of physical measurement, we introduce an *a posteriori* error estimator based on the evaluation of the residual of the temperature equation. For a given parameter μ , the residual is defined as

$$\mathcal{R}(\mu) = \partial_t T + \mathbf{u} \cdot \nabla T - \frac{1}{\text{Re Pr}} \nabla^2 T + S(T), \quad \text{in } \Omega, \quad (8)$$

$$\mathcal{R}(\mu) = T - \mu, \quad \text{on } \Gamma_l, \quad (9)$$

with $\mathbf{u} = \mathbf{0}$ in the conductive regime, where Γ_l stands to the left boundary wall. This residual provides a quantitative measure of how well the POD-RBF approximation satisfies the underlying physical model and boundary conditions. Unlike purely data-driven indicators, it balances interpolation accuracy with physical consistency and serves as a reliable error surrogate. Rather than constructing the reduced-order model using all parameters simultaneously, we adopt a Greedy strategy that iteratively enriches the training set by selecting the most informative parameters according to the residual magnitude. Starting from an initial reduced basis built using a small number of parameters, the algorithm successively identifies the parameter for which the current surrogate model exhibits the largest residual and augments the snapshot set accordingly. The POD-RBF Greedy procedure is summarized in the following:

Algorithm 1 POD-RBF Greedy algorithm

Require: Parameter set $\mathcal{P}$, residual tolerance TOL

- 1: Initialize training set $\mathcal{P}_{\text{train}} \leftarrow \{\mu_1, \mu_{N_s}\}$
 - 2: Compute snapshots and construct initial POD basis ϕ
 - 3: **while** $\max_{\mu \in \mathcal{P}} \mathcal{R}(\mu) > \text{TOL}$ **do**
 - 4: Construct RBF weights w
 - 5: Evaluate $\mathcal{R}(\mu)$ for all $\mu \in \mathcal{P}$ (in parallel)
 - 6: Select

$$\mu^* = \arg \max_{\mu \in \mathcal{P}} \mathcal{R}(\mu)$$
 - 7: Update $\mathcal{P}_{\text{train}} \leftarrow \mathcal{P}_{\text{train}} \cup \{\mu^*\}$
 - 8: Update ϕ
 - 9: **end while**
 - 10: **return** ϕ, w
-

After the training stage, the reduced basis and the RBF interpolation weights can be stored and reused efficiently for online evaluations. Moreover, the Greedy procedure enables straightforward retraining or adaptive enrichment when new parameter values become available. To valid our POD-RBF interpolation method, we consider the POD-Glerkin approach as baseline which is described in Appendix A.

4 Numerical results

In this section, we assess the performance of the proposed POD–RBF reduced-order modeling strategy on both melting PCM configurations introduced in Sections (1) and (3), corresponding respectively to the purely conductive and convective regimes. The full-order models (FOM) are constructed using a conventional finite element method, following the implementation provided in the PCM numerical toolbox developed in [SRT⁺20]. All simulation scripts and associated source files are openly available at [PCM_Toolbox.DDM.2D](#).

Let V_h denote the finite element space spanned by continuous $\mathbb{P}_2$ polynomial basis functions. The notation $(\cdot, \cdot)$ refers to the weighted L^2 inner product, while ∂_t and ∇_h represent the discrete time derivative and the discrete gradient operator, respectively. High-fidelity snapshots are generated for several values of the hot-wall temperature parameter $\mu \in [1, 1.1]$. Each simulation is carried out up to a final time $t_{\max} = 51.6$ using a uniform time step $\delta t = 0.5$, resulting in $N_t = 100$ total of time steps per trajectory.

4.1 PCM in conductive regime

The weak formulation of the equation (1) reads:

$$(\partial_t T_h^{n+1}, v) + \frac{1}{\text{Re Pr}} (\nabla_h T_h^{n+1}, \nabla_h v) + (\partial_t S(T_h^{n+1}), v), \quad v \in V_h. \quad (10)$$

The left boundary of the domain is heated and maintained at a non-dimensional temperature $T = \mu$, above the fusion temperature $T_f = 0$, while the remainder of the domain is initially cold at $T_0 = -0.01$, slightly below the fusion point. This configuration ensures that the material is initially solid, with melting initiated at the heated wall. Heat diffusion from the boundary drives the phase-change process, producing a melting front that propagates into the cold region. Figures 2 shows more the heat transfer from left to the right wall.

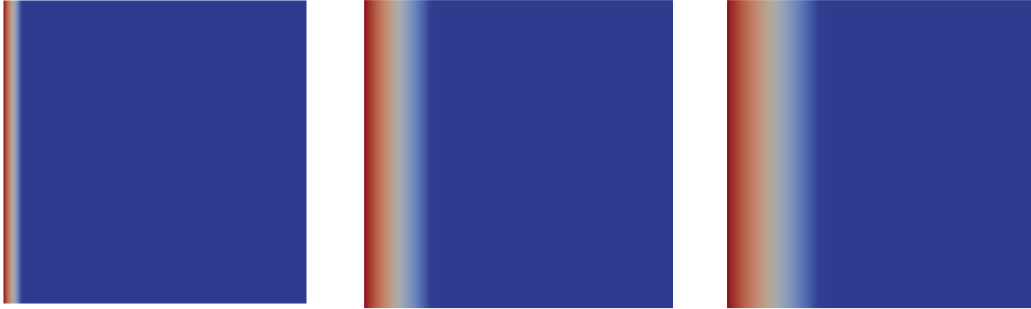


Figure 2: Conductive regime: Snapshot of the reference temperature at different times

Once the reduced solution T_r is built, the following quantities are evaluated on the reference finite element:

$$\mathcal{E}_2^2 = \frac{1}{N_t} \sum_{n=1}^{N_t} \frac{(T_h^n - T_r^n, T_h^n - T_r^n)}{(T_h^n, T_h^n)},$$

$$\mathcal{R}_2^2 = \frac{1}{N_t} \sum_{n=1}^{N_t} (R_r^n, R_r^n),$$

where

$$(R_r^n, v) = (\partial_t T_r^{n+1}, v) + \frac{1}{\text{Re Pr}} (\nabla_h T_r^{n+1}, \nabla_h v) + (S(T_r^{n+1}), v), \quad v \in V_h.$$

Figure 3 illustrates the evolution of the relative error, the corresponding residual and the reduced dimension r , as functions of the energy thresholds ϵ . Clearly we observe that as ϵ decreases, more POD modes are retained, leading to an increase dimension r and a corresponding improvement accuracy

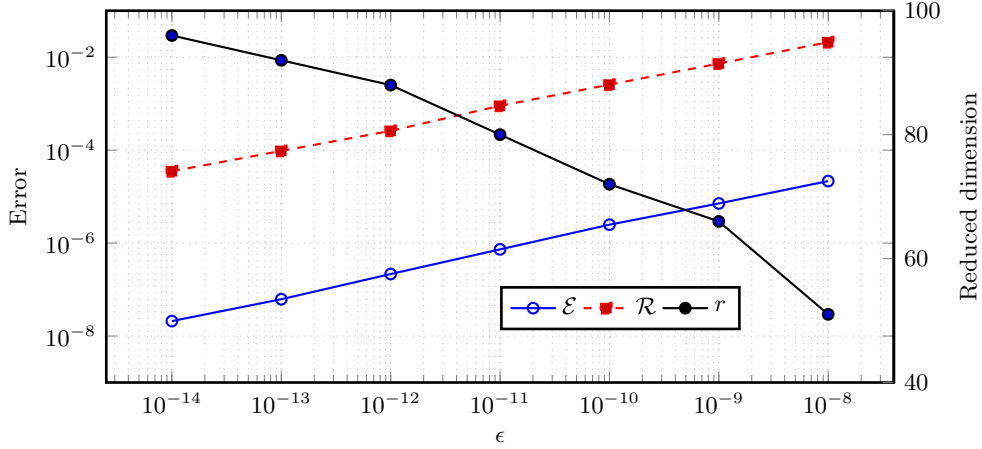


Figure 3: Conductive regime: Error and residuals resulting to the ROM model for $\mu = 1^\circ\text{C}$.

for the ROM. In particular, for sufficiently small values of ϵ , both the relative error and the residual drop below 10^{-4} . This behavior is expected, since the POD approximation converges to the full-order model as the truncated modes increases. Table 1 provides a quantitative comparison between the

Table 1: Conductive regime: approximation errors for POD-RBF and POD-Galerkin.

ϵ	POD-Galerkin			POD-RBF		
	r	$\mathcal{E}_2$	$\mathcal{R}_2$	r	$\mathcal{E}_2$	$\mathcal{R}_2$
10^{-8}	44	3.8×10^{-5}	0.031	51	2.2×10^{-5}	0.021
10^{-9}	64	1.7×10^{-5}	0.007	66	7.1×10^{-6}	0.007
10^{-10}	71	6.1×10^{-6}	0.002	72	2.5×10^{-6}	0.002

proposed POD-RBF model and the intrusive POD-Galerkin projection in terms of approximation errors, residual indicators, and reduced dimensions. For all tolerances ϵ , POD-Galerkin modeling holds less truncated modes than POD-RBF which is coherent to the fact that the reduced basis of POD-Galerkin results from L^2 scalar product while POD-RBF employs non geometric SVD where the mass matrix is equal to the identity. As the energy tolerance is tightened from 10^{-8} to 10^{-10} , both models improve as expected, but POD-RBF exhibits systematically lower state errors, reaching errors of order 10^{-6} while maintaining residuals of order 10^{-3} . From a computational point of view, the difference is even more significant. The intrusive POD-Galerkin model requires solving a nonlinear reduced system at each time step through Newton iterations, leading to online computational costs more than one hour in the present simulations. By contrast, the POD-RBF surrogate evaluates the solution through interpolation of reduced coefficients and reconstructs the field in only a few minutes. This substantial speedup, combined with improved accuracy, highlights the advantage of the proposed POD-RBF framework for fast parametric simulation problems.

Next we enhance the previous POD-RBF model to multiple values of μ following the Greedy iterative Algorithm 1. In this purpose, we introduce an a posteriori error estimator based on the residual quantity R_2 defined above. Indeed, optimizing R_2 simultaneously reduces the approximation error $\mathcal{E}$, as illustrated in Figure 3. The energy tolerance is fixed to $\epsilon = 10^{-10}$ and the training interval $[1, 1.1]$ is uniformly partitioned into $N_p = 9$ points, forming the parameter set

$$\mathcal{T}_\mu = \{1, 1.015, 1.03, 1.045, 1.06, 1.07, \dots, 1.09, 1.1\}.$$

Figure 4 reports the evolution of the error indicators throughout the Greedy iterations. At the eighth iteration, the prescribed residual tolerance is reached. At this stage, the reduced model retains $r = 104$ modes for the classical (batch) POD and $r = 87$ modes for the incremental POD approach. Both values are significantly smaller than the total number of snapshots $N_s = 900$ and drastically reduced compared to the full-order finite element dimension $N_h = 14641$. Moreover, it is observed

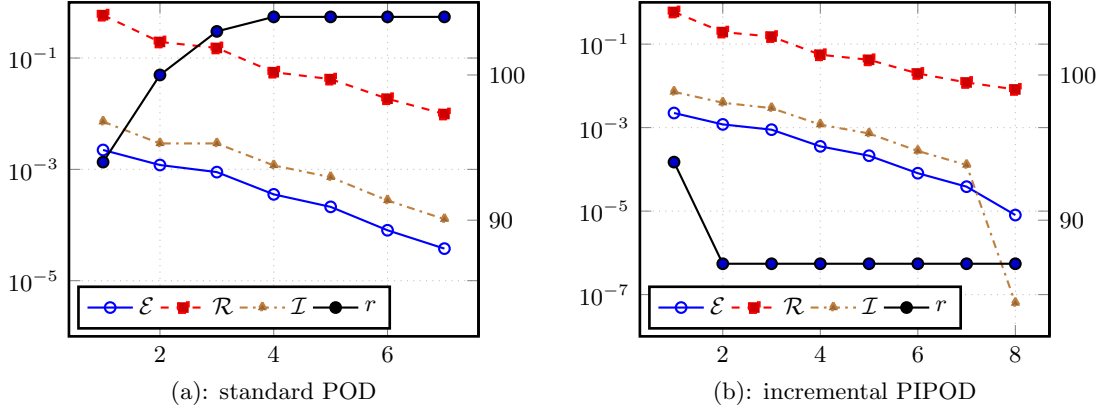


Figure 4: Conductive regime: Greedy iteration - error indicators (left) - reduced basis dimension r (right).

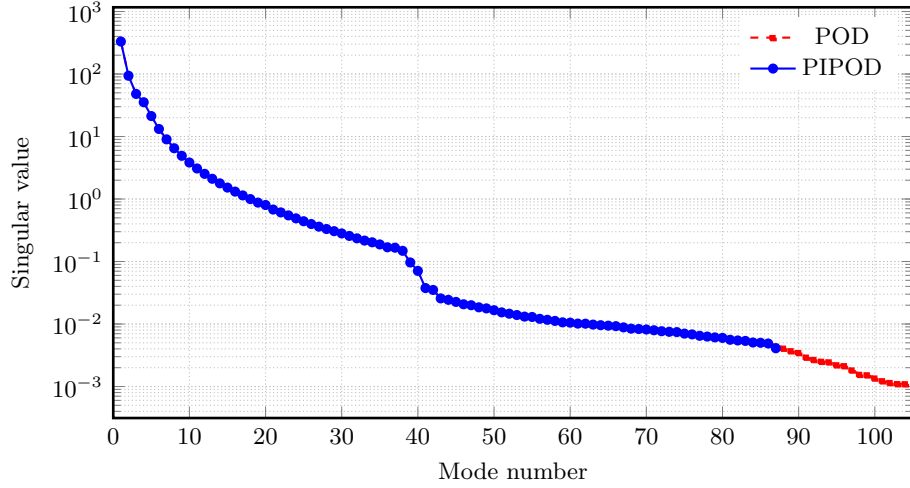


Figure 5: Singular values obtained with classical POD and Incremental PIPOD.

that the incremental POD variant achieves a more compact reduced basis without degrading either the global approximation accuracy or the interpolation error. This highlights the efficiency of the incremental strategy in controlling the growth of the reduced space. Figure 5 further confirms this observation, showing that the singular values obtained with incremental POD closely match those of the classical batch POD. This demonstrates that the proposed incremental procedure successfully preserves the dominant spectral content of the snapshot matrix.

Regarding to the training time, the table below summarizes the cost of both strategies in offline phase. The serial training cost of PIPOD is comparable to that of classical POD, confirming the efficiency

Table 2: Offline time comparison for POD and PIPOD with and without parallelization.

Method	Parallelization	Training time	Speedup
POD	Serial	1159.81	1
POD	OpenMP	304.08	3.82
PIPOD	Serial	1188.02	1
PIPOD	OpenMP	324.97	3.65

of the incremental formulation. When OpenMP parallelization is enabled, both approaches benefit significantly, achieving speedups of approximately 3.8 for POD and 3.6 for PIPOD. The slightly lower parallel efficiency of PIPOD is attributed to its sequential basis enrichment and repeated low-rank

updates.

The evolution of the errors and residuals associated with the selected target parameters throughout the iterative process as well as the RBF interpolation $\mathcal{I}_2$ given by:

$$\mathcal{I}_2^2(\mu^*) = \frac{1}{N_t} \sum_{n=1}^{N_t} \frac{\sum_{i=1}^r |\alpha_i(\mu^*, t_n) - (X_h(\mathbf{x}, t_n; \mu^*), \phi_i)|^2}{\sum_{i=1}^r |(X_h(\mathbf{x}, t_n; \mu^*), \phi_i)|^2}$$

are illustrated in Figure 4. Table 3 reports the interpolation and residual errors obtained for two new

Table 3: Conductive regime: Interpolation errors for the RBF, POD-RBF and POD-Galerkin methods.

μ	POD-RBF			POD-Galerkin		
	$\mathcal{I}_2$	$\mathcal{E}_2$	$\mathcal{R}_2$	$\mathcal{I}_2$	$\mathcal{E}_2$	$\mathcal{R}_2$
1.035	4.6×10^{-5}	1.4×10^{-5}	0.004	5.7×10^{-5}	1.4×10^{-5}	0.005
1.075	7.8×10^{-5}	2.3×10^{-5}	0.007	6×10^{-5}	1.4×10^{-5}	0.006

parameters, $\mu = 1.035$ and $\mu = 1.075$, lying inside the training interval. The POD-Galerkin uses the same training parameters as for the POD-RBF with fixed reduced dimension $r = 87$. The results clearly highlight once again the advantages of the proposed POD-RBF approach which holds the same error orders as POD-Galerkin. More importantly, the residual error $\mathcal{R}_2$ associated with the POD-RBF model remains very small, below the prescribed residual tolerance 10^{-2} for both test parameters. This indicates that the reduced-order solution not only interpolates the data accurately but also satisfies the governing heat equation to a high degree of consistency. The low residual confirms that the essential physics of heat diffusion and phase change are preserved by the reduced model. Overall, these results demonstrate that the POD-RBF method provides a robust and accurate non-intrusive reduced-order model for conductive phase-change problems, while maintaining excellent physical consistency. In

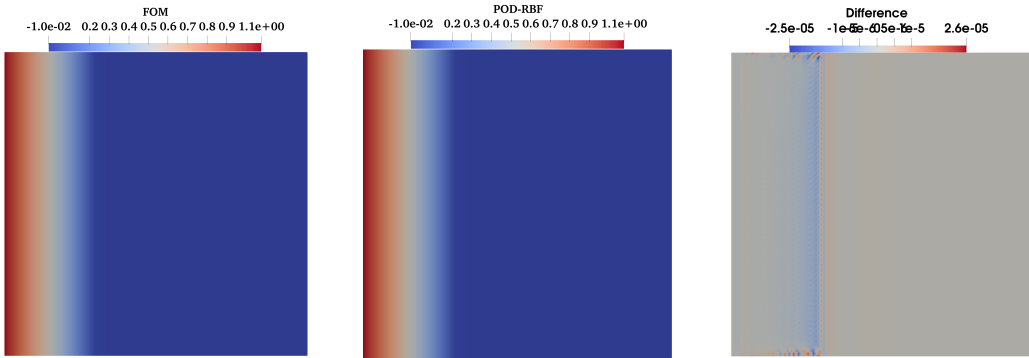


Figure 6: Conductive regime: Comparison between reference temperature and interpolated T_r at time $t = 51.6$ for $\mu = 1.075^\circ C$.

addition, the results plotted in Figure 6 demonstrate a very good agreement between the high-fidelity temperature field and the interpolated POD-RBF solution at the final time $t = 51.6$. For parameter $\mu = 1.075$, the reduced-order model accurately reproduces the spatial structure of the thermal field, including the location and shape of the melting front. This visual consistency is in full agreement with the small interpolation and residual errors reported in Table 3, and it validates the ability of the POD-RBF approach to generalize reliably to new parameter values in the conductive regime.

4.2 PCM in convective regime

After analyzing the PCM in the purely conductive regime, we now extend the study to a more realistic configuration in which convection plays a dominant role in the melting dynamics. In this regime, temperature gradients induce buoyancy forces that generate fluid motion within the molten phase, significantly enhancing heat transfer and altering the evolution of the phase interface. As a result,

the coupling between momentum and energy equations introduces strong nonlinearities and multiscale dynamics, making the numerical simulation considerably more challenging than in the conduction-dominated case. The POD–RBF model is here assessed in its ability to approximate the coupled velocity–pressure–temperature dynamics. Special attention is devoted to evaluating the accuracy and robustness of the reduced model through a posteriori residual-based error estimators, as well as its predictive capability for previously unseen parameter values.

We introduce first the functions:

$$f(T) = \frac{\text{Ra} T}{\text{Pr Re}^2}, \quad \chi(T) = \frac{C_{ck}(1 - \zeta_l(T))^2}{\zeta_l(T)^3 + 10^{-6}}.$$

Our reference solution in full order dimension satisfies the weak formulation:

$$\begin{aligned} (\nabla_h \cdot \mathbf{u}_h^{n+1}, q) &= 0, \quad \forall q \in W_h, \\ (\partial_t T_h^{n+1} + \mathbf{u}_h^{n+1} \cdot \nabla_h T_h^{n+1}, v) + \frac{1}{\text{Re Pr}} (\nabla_h T_h^{n+1}, \nabla_h v) + (\partial_t S(T_h^{n+1}), v) &= 0, \quad \forall v \in V_h, \\ (\partial_t \mathbf{u}_h^{n+1}, \mathbf{v}) + ((\mathbf{u}_h^{n+1} \cdot \nabla) \mathbf{u}_h^{n+1}, \mathbf{v}) + \frac{1}{\text{Re}} (\nabla_h \mathbf{u}_h^{n+1}, \nabla_h \mathbf{v}) - (p_h^{n+1}, \nabla_h \cdot \mathbf{v}) \\ &\quad - (f(T_h^{n+1}) \mathbf{e}_y + \chi(T_h^{n+1}) \mathbf{u}_h^{n+1}, \Psi) = 0, \quad \forall \mathbf{v} \in V_h^2 \end{aligned}$$

The numerical experiments are carried out with the Taylor–Hood finite elements on a 60×60 square mesh. Initially, the temperature is set to $T = -0.01$, and Figure 7 illustrates the results for several values of μ at the final time $t_{\max} = 51.6$.

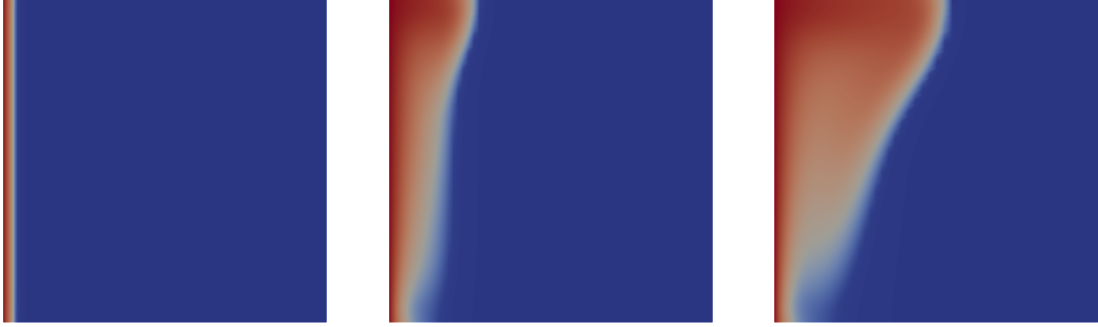


Figure 7: Convective regime: Reference temperature for $\mu = 1^\circ\text{C}$ at different times

As in the previous test case, we consider the residual (8) discretized by $\mathcal{R}_2^2 = \delta t \sum_{n=1}^{N_t} (\mathcal{R}_r^n, \mathcal{R}_r^n)$ where

$$(\partial_t \mathcal{R}_r^{n+1}, v) = (\partial_t T_r^{n+1} + \mathbf{u}_r^{n+1} \cdot \nabla_h T_r^{n+1}, v) + \frac{1}{\text{Re Pr}} (\nabla T_r^{n+1}, \nabla v) + (S(T_r^{n+1}), v), \quad v \in V_h. \quad (11)$$

The definition of the relative error $\mathcal{E}_2$ and the energy residual remain unchanged. The model is trained over the parameter set $\mu \in \mathcal{T}_\mu = \{1., 1.01, 1.02, \dots, 1.09, 1.1\}$, with each parameter corresponding to 50 time steps. Again the reduced model is initially constructed using the lower and upper parameters and then iteratively optimized through the Greedy Algorithm 1 with a tolerance $\text{TOL} = 0.1$ and an energy threshold $\epsilon = 10^{-9}$. Figure 8 illustrates the convergence behavior of the POD–RBF model after ten iterations. The evolution of the error indicators and the reduced basis dimension is reported for both the standard POD Greedy strategy (panel (a)) and the PIPOD Greedy variant (panel (b)). For both approaches, the relative error, the residual-based a posteriori estimator, and the interpolation error exhibit early a slow decay as the Greedy iterations progress before reached the prescribed tolerance at the last iteration. This trend confirms that the adaptive enrichment of the training set effectively improves the accuracy of the ROM by progressively capturing the most energetic and dynamically relevant features of the convective flow. The convergence of all error indicators below the prescribed tolerance confirms the reliability of the resulting POD–RBF model for parametric predictions. In the other hand, comparing panels (a) and (b), the PIPOD Greedy strategy achieves a comparable reduction in error indicators while maintaining a significantly smaller reduced basis dimension. In particular

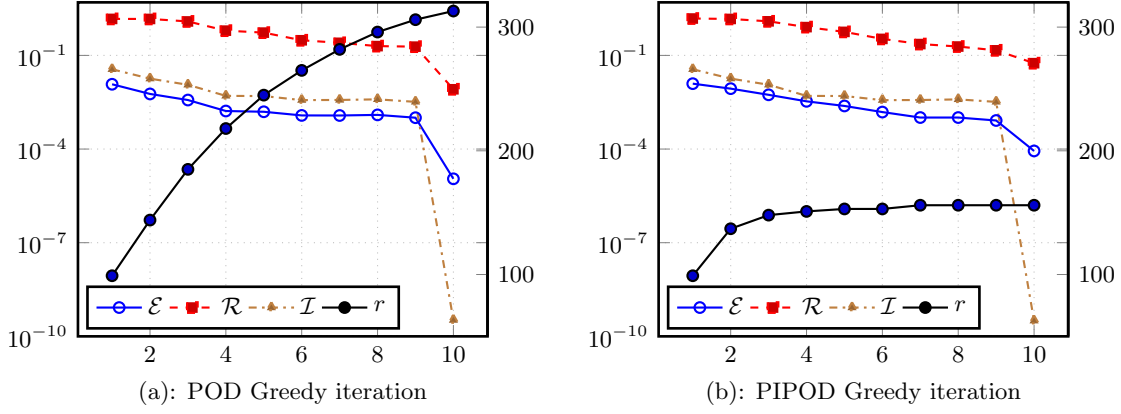


Figure 8: Convective regime: Greedy iteration - error indicators (left) - reduced basis dimension r (right).

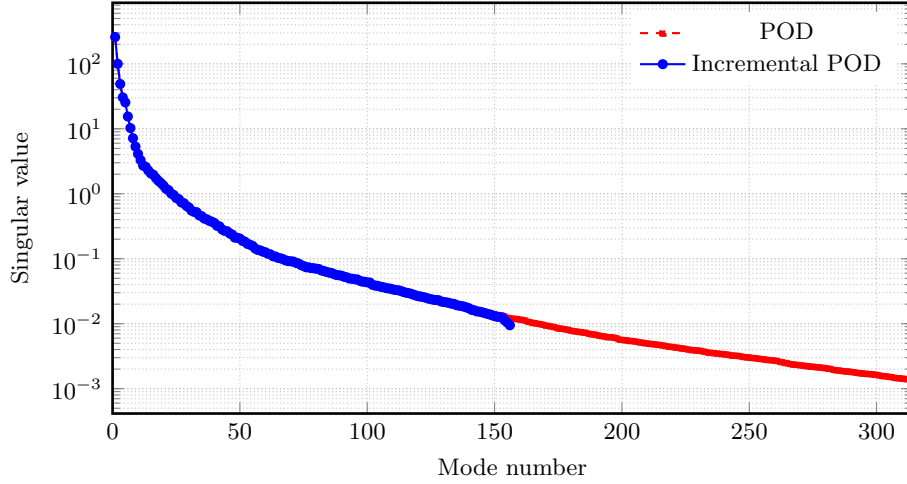


Figure 9: Convective regime: Singular values obtained with classical POD and Incremental POD.

Table 4: Convective regime: reduced basis obtained for velocity and temperature fields using POD and PIPOD.

Variable	POD modes	PIPOD modes	Rate (%)
u_1	472	338	71.6
u_2	465	309	66.5
T	313	156	49.8

as shown in Figure 9, the standard POD Greedy requires up to 313 modes to reach the prescribed tolerance, whereas the incremental POD approach stabilizes around $r = 156$.

Furthermore Table 4 reports the history of the reduced basis for the velocity components and temperature. Here the rate represents the ratio between the number of modes retained by PIPOD and standard POD. Results show that PIPOD achieves a significant reduction of the reduced basis dimension, particularly for the reduced temperature field where nearly 50% fewer modes are required. This demonstrates the ability of the incremental POD technique to reuse previously computed modes efficiently and limit unnecessary growth of the reduced space.

We report also the offline computational cost associated with the construction of the POD-RBF model using both POD and PIPOD strategies. As shown in Table 5, PIPOD requires a larger offline cost than standard POD due to the incremental basis updates performed during the greedy procedure as seen in the conductive regime. Nevertheless, OpenMP parallelization significantly reduces the training time

Table 5: Convective regime: Offline training time comparison for POD-RBF using POD and PIPOD strategies, with and without OpenMP parallelization.

Method	Parallelization	Training time (s)	Speedup
POD	Serial	689.56s	1
POD	OpenMP	271.84	2.54
PIPOD	Serial	1047.78	1
PIPOD	OpenMP	611.09	1.71

for both methods. In particular, a speedup of approximately 2.5 is achieved for POD, while PIPOD benefits from a speedup of about 1.7.

Table 6: Convective regime: Comparison between POD-RBF and linear interpolation.

ϵ	POD-RBF			POD-Galerkin	
	$\mathcal{I}_2$	$\mathcal{E}_2$	$\mathcal{R}_2$	$\mathcal{E}_2$	$\mathcal{R}_2$
1.035	1.5×10^{-3}	4.7×10^{-4}	0.047	3.8×10^{-5}	0.031
1.075	1.3×10^{-3}	4.4×10^{-4}	0.047	1.7×10^{-5}	0.007

We now evaluate the resulting model for the parameters $\mu = 1.035$ and $\mu = 1.075$, which were not included in the training set. Since the velocity field requires a significantly larger number of POD modes to be accurately reconstructed, as shown in Table 4, a full POD-Galerkin reduction of the coupled velocity–pressure–temperature system would lead to a considerably larger reduced dimension and computational cost. To facilitate a fair comparison between the POD-RBF and conventional POD-Galerkin approaches, only the temperature field is reduced, while the high-fidelity velocity snapshots are employed for the evaluation of the residual error estimator (11). Table 6 compares the proposed POD-RBF reduced-order model, and hybrid POD-Galerkin at time $t = 25.6$. The residual-based error indicator $\mathcal{R}$ highlights a clear distinction between the two interpolations. While both POD-RBF and linear achieve comparable interpolation errors $\mathcal{E}$, the POD-RBF model yields substantially smaller residuals. For instance, at $\mu = 1.035$, the residual error is reduced from 9.710^{-2} (linear) to 4.710^{-2} (POD-RBF), and a similar reduction is observed at $\mu = 1.075$. This confirms that the POD-RBF model better satisfies the underlying governing equations and is consistent with the physical constraints. Furthermore Figure 10 below shows better agreement between the POD-RBF interpolation and the high fidelity FOM solution at the final time.

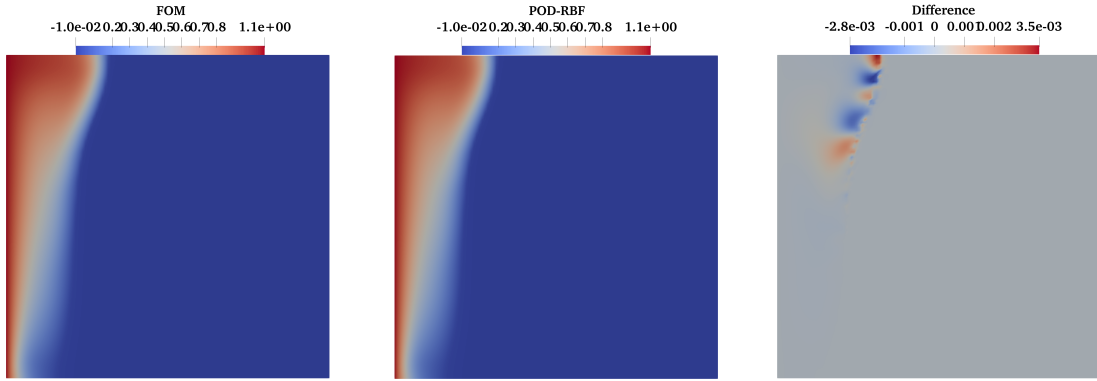


Figure 10: Convective regime: Comparison between FOM and POD-RBF temperatures at time $t = 25.6$ for $\mu = 1.075^\circ\text{C}$.

5 Conclusion

In this paper, we investigated POD methods coupled with a non-intrusive RBF interpolation strategy for the efficient simulation of phase-change material problems, encompassing both purely conductive and fully convective regimes. A qualitative comparison was performed against the conventional, intrusive POD-Galerkin method as a reference ROM. Two POD variants were evaluated: the classical batch POD and a pseudo-incremental POD approach, the latter designed to reduce the number of retained modes while maintaining high accuracy.

The primary contribution of the proposed POD–RBF framework is the incorporation of an *a posteriori* residual-based error estimator. This estimator provides a physically consistent indicator of model accuracy by directly quantifying how well the reduced solution satisfies the governing equations. Beyond simply assessing statistical interpolation quality, it identifies regions within the parameter space where the reduced basis becomes insufficient. When embedded within a Greedy sampling algorithm, the estimator successfully guides the adaptive selection of the most informative parameter instances. This results in a highly efficient offline training process, enhanced global generalization accuracy, and a significant reduction in the total number of high-fidelity snapshots required. Consequently, this adaptive strategy ensures a robust balance between computational efficiency and approximation quality, particularly in strongly nonlinear regimes.

Overall, the proposed POD–RBF framework demonstrates strong performance in both of the studied configurations. In the conductive regime, despite the presence of discontinuous enthalpy terms in the energy equation, the POD-RBF framework achieves high accuracy and yields approximation and residual errors of the same order of magnitude as the POD-Galerkin method. This confirms the robustness of the non-intrusive ROM approach for tracking sharp phase-transition fronts. Conversely, in the convective regime, the reduction process becomes more challenging due to the severe nonlinearities introduced by the Navier–Stokes dynamics and the mushy-zone penalization technique. In particular, the accurate reconstruction of the velocity field necessitates a larger number of POD modes (see Table 4), ultimately leading to higher reduced dimensions.

Future developments will pursue two distinct trajectories tailored to the specific physics of each PCM regime. For the conductive regime, we will focus on enhancing the RBF interpolation stage with a physics-informed correction mechanism regularized by the model residual. To minimize computational overhead, a dynamic switching strategy can be introduced: the standard non-intrusive RBF prediction is used directly if it satisfies a physical tolerance, whereas the optimization-based correction is activated only when the residual exceeds this threshold. This strategy is formulated as follows:

$$\begin{aligned} \tilde{a}_r(t, \mu) &= \mathcal{I}_{\text{RBF}}^r(t, \mu), \\ a_r(t, \mu) &= \begin{cases} \tilde{a}_r(t, \mu), & \text{if } \|\mathcal{R}(\tilde{a}_r)\|^2 \leq \text{TOL}, \\ \arg \min_a (\|a - \tilde{a}_r(t, \mu)\|^2 + \lambda \|\mathcal{R}(a)\|^2), & \text{if } \|\mathcal{R}(\tilde{a}_r)\|^2 > \text{TOL}, \end{cases} \end{aligned}$$

where $\mathcal{R}(\cdot)$ denotes the model residual, TOL is a user-defined physical tolerance, and $\lambda > 0$ is a regularization parameter. Resolving this optimization problem when the threshold is violated presents a significant numerical challenge because, as detailed in the preceding sections, the residual depends highly nonlinearly on the reduced coefficients. This physics-informed non-intrusive ROM feedback mechanism could be deployed during both the offline and online stages to ultimately stabilize long-term time extrapolation.

Conversely, for the convective regime, where linear POD exhibits inherent limitations due to advection-dominated dynamics, we will investigate methods capable of handling moving features and shifting phase boundaries. Specifically, we will explore localized or piecewise reduced basis methods [PBWB14], utilizing cluster-based partitioning techniques [KNC⁺14] to segment the system’s states (e.g., solid, liquid, and mushy zones) into distinct local subregions. This allows the model to dynamically switch between highly specialized, low-dimensional local spaces rather than relying on a massive global basis. Alternatively, we will investigate kinematic approaches such as shifted POD frameworks [RSSM18] and fully nonlinear manifold learning techniques via convolutional autoencoders [FDQ21]. This transition from a traditional linear global subspace to an adaptive, localized, or advection-aware reduced manifold will allow the surrogate model to efficiently track the sharp, moving phase-change interface while keeping the online reduced dimensions strictly minimized.

Funding

This work is financially supported by the French National Research Agency (ANR SINRAM - Project-ANR-23-ASTR-0003).

A POD-Galerkin model

We review in this section the ROM Galerkin approach applied to PCM problem and for simplicity we present the approach for the conductive regime only. The ROM based on the Galerkin approach consists in projecting the full-order problem (10) onto the reduced space $V_r = \text{span}\{\phi_1, \dots, \phi_r\}$. Then the reduced problem reads: find $T_r \in V_r$ such that

$$c_1(T_r^{n+1} + S(T_r^{n+1}), \phi_i) + \frac{1}{\text{Re Pr}}(\nabla T_r^{n+1}, \nabla \phi_i) = (c_2(T_r^n + S(T_r^n)) - c_3(T_r^{n-1} + S(T_r^{n-1})), \phi_i), \quad \forall \phi_i \in V_r. \quad (12)$$

The reduced basis $\{\phi_i\}_{i=1}^r$ is constructed using POD from a set of snapshots $X_h = [T_h^1, \dots, T_h^N]$. After centering the snapshots

$$\tilde{X}_h^k = T_h^k - \bar{X}_h, \quad \text{with } \bar{X}_h = \frac{1}{N} \sum_{k=1}^N T_h^k,$$

the correlation matrix is defined as

$$C_{ij} = (\tilde{X}_h^i, \tilde{X}_h^j), \quad 1 \leq i, j \leq N.$$

Let (λ_i, v_i) be the elements of C , sorted in descending order. The POD modes are then given by

$$\phi_i = \frac{1}{\sqrt{\lambda_i}} \tilde{X}_h v_i, \quad i = 1, \dots, r.$$

To ensure an orthonormal basis satisfying $(\phi_i, \phi_j) = \delta_{i,j}$, careful selection of the truncated dimension r is required. In practice, this is achieved using the energy criterion based on a prescribed tolerance ϵ . Thus reduced temperature is expressed as

$$T_r^{n+1} = \bar{X}_h + \sum_{i=1}^r \alpha_i^{n+1} \phi_i.$$

Substituting this expansion into (12) leads to the following nonlinear system for the coefficients α_i^{n+1} , solved by Newton-type iterative procedure:

$$\begin{aligned} \sum_{j=1}^r \left[c_1(\phi_j + S'(T_r^k) \phi_j, \phi_i) + \frac{1}{\text{Re Pr}}(\nabla \phi_j, \nabla \phi_i) \right] \alpha_j^{k+1} = & c_1(S'(T_r^k) T_r^k - S(T_r^k) - S'(T_r^k) \bar{X}_h, \phi_i) \\ & + (c_2(T_r^n + S(T_r^n)) - c_3(T_r^{n-1} + S(T_r^{n-1})), \phi_i) \\ & - c_1(\bar{X}_h, \phi_i) - \frac{1}{\text{Re Pr}}(\nabla \bar{X}_h, \nabla \phi_i) \end{aligned}$$

B POD and PIPOD algorithms

Algorithm 2 POD

Require: $\{u(\mu_i, t_j)\}_{1 \leq i \leq N_s, 1 \leq j \leq N_t} \subset \mathbb{R}^{N_h}, \varepsilon$

Ensure: $\Phi \in \mathbb{R}^{N_h \times r}, \theta \in \mathbb{R}^{r \times M}, M = N_s \times N_t$

- 1: $X = [u_1, \dots, u_M] \in \mathbb{R}^{N_h \times M}$
 - 2: $\tilde{X} = X^T - \frac{1}{M} \sum_{k=1}^M u_k$
 - 3: $\tilde{X} = V \Sigma U^T$
 - 4: Determine r such that $E_r \geq 1 - \varepsilon$
 - 5: $\Phi_i = \frac{1}{\Sigma_{i,i}} \tilde{X}^T V_i, \quad i = 1, \dots, r.$
 - 6: $\theta = \Phi^T \tilde{X}$
 - 7: **return** Φ, θ
-

Algorithm 3 PIPOD

Require: Φ, Σ for initial parameter μ_1 , tolerance ε_p

Ensure: Updated basis Φ for new parameter μ^*

- 1: **for** each new snapshot $\tilde{u}(\mu^*, t_j)$ **do**
- 2: $p = \Phi^T \tilde{u}$
- 3: $r = \tilde{u} - \Phi p$
- 4: **if** $\|r\| \leq \varepsilon_p \|\tilde{u}\|$ **then**
- 5: **break**
- 6: **end if**
- 7:

$$K = \begin{bmatrix} \Sigma & p \\ 0 & \|r\| \end{bmatrix}$$

- 8: $K = \hat{U} \hat{\Sigma} \hat{V}^T$
- 9: Update basis:

$$\Phi \leftarrow [\Phi_0 \quad \frac{r}{\|r\|}] \hat{U}_r$$

- 10: **end for**
 - 11: **return** $\Phi, \hat{\Sigma}$
-

References

- [AGZ⁺15] Icíar Alfaro, David González, Sergio Zlotnik, Pedro Díez, and Francisco Chinesta. An error estimator for real-time simulators based on model order reduction. *Advanced Modeling and Simulation in Engineering Sciences*, 2:30, 2015.
- [BGW15] P. Benner, S. Gugercin, and K. Willcox. A survey of projection-based model reduction methods. *SIAM Review*, 2015.
- [Buh03] M. D. Buhmann. *Radial Basis Functions: Theory and Implementations*. Cambridge University Press, 2003.
- [BVR88] A. D. Brent, V. R. Voller, and K. J. Reid. Enthalpy-porosity technique for modeling convection-diffusion phase change. *Numerical Heat Transfer*, 1988.
- [CS10] S. Chaturantabut and D. C. Sorensen. Nonlinear model reduction via discrete empirical interpolation. *SIAM J. Sci. Comput.*, 2010.
- [CSZX21] P. Cheng, A. S. Soliman, S. Zhu, and L. Xu. Numerical simulation and experimental verification of constrained melting of phase change materials in a cylindrical enclosure. *Journal of Energy Storage*, 35:102268, 2021.

- [FCB24] Lihong Feng, Sridhar Chellappa, and Peter Benner. A posteriori error estimation for model order reduction of parametric systems. *Advanced Modeling and Simulation in Engineering Sciences*, 11, 2024.
- [FDQ21] Stefania Fresca, Luca Dede’, and Alfio Quarteroni. A comprehensive deep learning-based approach to reduced order modeling of nonlinear time-dependent parametrized pdes. *Journal of Scientific Computing*, 87(1):1–36, 2021.
- [FKRAH04] M. M. Farid, A. M. Khudhair, S. A. K. Razack, and S. Al-Hallaj. A review on phase change energy storage: materials and applications. *Energy Conversion and Management*, 45:1597–1615, 2004.
- [GH18] M. Guo and J. S. Hesthaven. Reduced order modeling for nonlinear structural analysis using gaussian radial basis functions. *International Journal for Numerical Methods in Engineering*, 113(9):1450–1470, 2018.
- [GV96] C. Gau and R. Viskanta. Melting and solidification of a pure metal on a vertical wall. *Journal of Heat Transfer*, 1996.
- [HLB96] P. Holmes, J. Lumley, and G. Berkooz. *Turbulence, Coherent Structures, Dynamical Systems and Symmetry*. Cambridge University Press, 1996.
- [HRS16] J. S. Hesthaven, G. Rozza, and B. Stamm. Certified reduced basis methods for parametrized partial differential equations. *SpringerBriefs in Mathematics*, 2016.
- [HRS18] J. S. Hesthaven, G. Rozza, and B. Stamm. *Certified Reduced Basis Methods for Parametrized PDEs*. Springer, 2018.
- [KNC⁺14] Eurika Kaiser, Bernd R Noack, Laurent Cordier, Andreas Spohn, Marc Segond, Markus Abel, Nathan Brunton, and Steven L Brunton. Cluster-based reduced-order modelling of transient features. *Journal of Fluid Mechanics*, 754:365–414, 2014.
- [KV01] K. Kunisch and S. Volkwein. Galerkin proper orthogonal decomposition methods for parabolic problems. *Numerische Mathematik*, 90:117–148, 2001.
- [MPR15] Yvon Maday, Anthony T. Patera, and Einar M. Rønquist. The reduced basis method for parametrized partial differential equations and applications. *Comptes Rendus Mathématique*, 353(2):141–145, 2015.
- [PBWB14] Benjamin Peherstorfer, Denese Butnaru, Karen Willcox, and Hans-Joachim Bungartz. Localized trajectory-directed 1d-basis generation for nonlinear structural dynamics. *Computer Methods in Applied Mechanics and Engineering*, 268:204–221, 2014.
- [PW16] B. Peherstorfer and K. Willcox. Data-driven operator inference for nonintrusive projection-based rom. *Computer Methods in Applied Mechanics and Engineering*, 2016.
- [QMN16] A. Quarteroni, A. Manzoni, and F. Negri. *Reduced Basis Methods for Partial Differential Equations*. Springer, 2016.
- [RDD19] A. Rakotondrandisa, I. Danaila, and L. Danaila. Numerical modelling of a melting-solidification cycle of a phase-change material with complete or partial melting. *International Journal of Heat and Fluid Flow*, 76:57–71, 2019.
- [RSSM18] Julius Reiss, Philipp Schulze, Jörn Sesterhenn, and Volker Mehrmann. The shifted proper orthogonal decomposition: A mode decomposition for multiple transport phenomena. *SIAM Journal on Scientific Computing*, 40(3):A1322–A1344, 2018.
- [SRT⁺20] Georges Sadaka, Aina Rakotondrandisa, Pierre-Henri Tournier, Francky Luddens, Corentin Lothodé, and Ionut Danaila. Parallel finite-element codes for the simulation of two-dimensional and three-dimensional solid-liquid phase-change systems with natural convection. *Computer Physics Communications*, 257(107492), 2020.

- [SSB⁺21] S. KELBIJ STAR, GIOVANNI STABILE, FRANCESCO BELLONI, GIANLUIGI ROZZA, and JORIS DEGROOTE. A novel iterative penalty method to enforce boundary conditions in finite volume pod-galerkin reduced order models for fluid dynamics problems. *Communications in Computational Physics*, 30(1):34–66, Apr. 2021.
- [STCB09] A. Sharma, V. V. Tyagi, C. R. Chen, and D. Buddhi. Review on thermal energy storage with phase change materials and applications. *Renewable and Sustainable Energy Reviews*, 13:318–345, 2009.
- [VC87] V. R. Voller and M. Cross. Accurate solutions of moving boundary problems using the enthalpy method. *International Journal of Heat and Mass Transfer*, 1987.
- [VKS25] Ajitanshu Vedrtam, Kishor Kalauni, and Nelson Soares. Experimentally validated transient enthalpy-based fem model for pcm-integrated building materials with advanced thermal and energy efficiency insights. *Case Studies in Thermal Engineering*, 72:106374, 2025.
- [Vol90] V. R. Voller. An overview of numerical methods for solving phase change problems. *Advances in Numerical Heat Transfer*, 1990.
- [Vol11] S. Volkwein. Model reduction using proper orthogonal decomposition. *lecture notes*, 2011.
- [WA19] Z. Wang and I. Akhtar. Pod-based reduced-order modeling with radial basis function interpolation for nonlinear systems. *Journal of Computational Physics*, 384:96–115, 2019.
- [WACRX25] Liang Wu, Mejdi Azaiez, Tomás Chacón Rebollo, and Chuanju Xu. Reduced basis method based on a posteriori error estimate for the parameterized allen–cahn equation. *Journal of Computational and Applied Mathematics*, 2025.
- [XFPH15] D. Xiao1, F. Fang, C. Pain, and G. Hu. Non-intrusive reduced order modelling of the navier-stokes equations based on rbf interpolation. *Int. J. Numer. Meth. Fluids*, 79(11):580–595, 2015.
- [YA24] H. Yao and M. Azaiez. A monolithic model of solid–liquid phase change problem. *Computer Methods in Applied Mechanics and Engineering*, 421:116794, 2024.
- [ZMCM03] B. Zalba, J. M. Marín, L. F. Cabeza, and H. Mehling. Review on thermal energy storage with phase change: materials, heat transfer analysis and applications. *Applied Thermal Engineering*, 23(3):251–283, 2003.